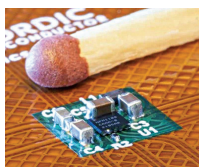


COMPONENTS

Power management IC

The nPM1100 combines a USB-compatible input regulator with overvoltage protection, 400mA battery charger, and 150mA DC-DC step-down voltage regulator in a



2.075×2.075 mm wafer level chip scale package. The PMIC ensures reliable power supply, maximised appli-

cation battery life and features an ultra-low quiescent current of 700nA. The product can also be used as a generic PMIC for any application using rechargeable lithium-ion or lithium-polymer batteries. Its extremely compact form factor (takes up as little as 23mm² of PCB area) makes it ideal for advanced wearables, connected medical devices, and other space-constrained applications.

Nordic Semiconductor
www.nordicsemi.com

Temperature and humidity sensor

A new sensor called Hadar Cold has been designed that can operate in the temperature range of -20°C to +60°C



by simply harvesting indoor lighting. The newly launched wireless, battery-free and maintenance-free temperature and humidity sensor includes

Epishine's solar cell, enabling energy harvesting in very low light levels. Benefits are sustainability and environmental friendliness. Now there is no worry about the operational costs involved in changing batteries. Applications include smart buildings, smart homes, and retail industry.

Pollux
www.pllx.se

Isolated gate driver

The Si828x version 2 isolated gate driver addresses the growing market for half- and full-bridge inverters and power supplies that require improved power density, cooler operation, and reduced switching losses. It includes features such as 4A peak gate



drive current, improved common mode transient immunity, undervoltage settings that improve flexibility for SiC FETs and guard against poorly regulated power supplies and eliminates parasitic-induced shoot-through conditions, offering a variety of benefits that make it ideal for hybrid and electric vehicles as well as industrial applications.

Silicon Labs
www.silabs.com

MPUs for AI applications

The RZ/V series of microprocessors (MPUs) incorporate Renesas' exclusive AI accelerator—the DRP-AI (dynamically reconfigurable processor) that makes embedded AI easier and more power efficient.



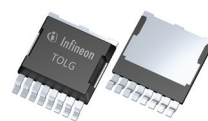
The new RZ/V2L has several features such as high-precision

AI inference capabilities combined with top-class power efficiency. It also includes optimisations such as tailoring the DRP-AI operating frequency and memory interface for an entry-level MPU. The DRP-AI provides both real-time AI inference and image processing functions that are essential for camera support such as colour correction and noise reduction, enabling development of AI based vision applications without requiring an external image signal processor.

Renesas Electronics Corporation
www.renesas.com

Power MOSFETs

Various types of electric vehicles, power tools, and battery management systems demand high current rating, ruggedness, and extended lifetime. Address-



ing the requirements for achieving maximum performance are two new OptiMOS

power MOSFET packages in the TOLx family: TOLG (TO-leaded with Gullwing leads) and TOLT (TO-leaded top-side cooling). Together, the TOLx family offers very low RdS(on) and a high current rating over 300A that increases system efficiency in high-power density designs. The main advantages of TOLG are in designs with aluminium-insulated metal substrate boards. In these designs, the coefficient of thermal expansion is higher than copper-IMS and FR4 boards.

Infiniteon Technologies AG
www.infineon.com

Space-qualified MOSFETs

Power supplies in space applications require enhanced radiation technology to withstand extreme particle



interactions and solar and electromagnetic events to avoid degradation of space based systems and disruption of operations. The M6 MRH25N12U3 radiation-hardened 250V, 0.21-ohm

Rds(on) MOSFET is suitable for commercial aerospace and defence space applications. The rad-hard MOSFET provides the primary switching element in power conversion circuits including point-of-load converters, DC-DC converters, motor drives and controls, and general-purpose switching. The device can withstand total ionising dose up to 100krad and 300krad and single event effects with linear energy transfer up to 87 MeV/mg/cm².

Microchip Technology
www.microchip.com